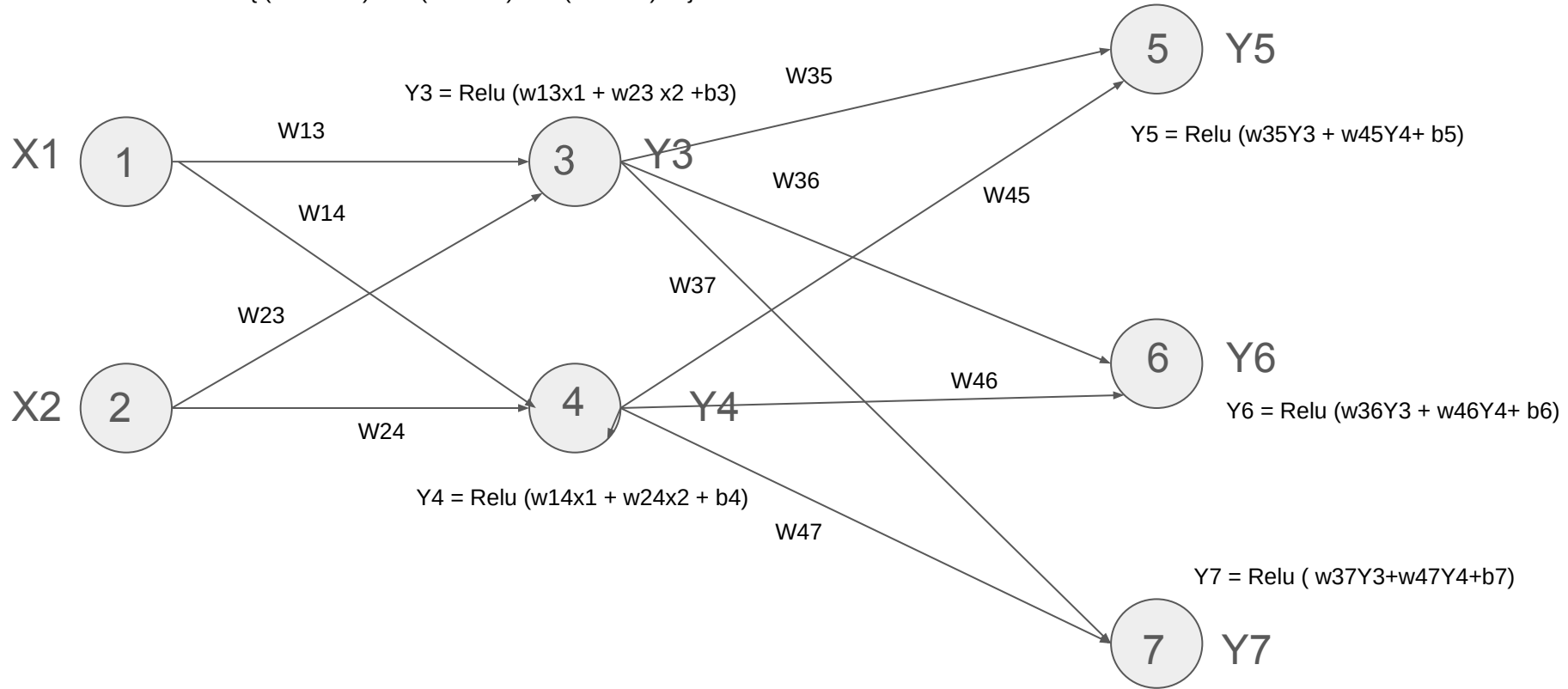


$$\text{Loss} = 1/2 \{ (\text{Ob5} - Y5)^2 + (\text{Ob6} - Y6)^2 + (\text{Ob7} - Y7)^2 \}$$



At first start with the output nodes. Define loss function and differentiate loss function with respect to all the output nodes.

$$\begin{aligned}
Loss &= \frac{1}{2} \{ (Ob_5 - Y_5)^2 + (Ob_6 - Y_6)^2 + (Ob_7 - Y_7)^2 \} \\
\frac{dLoss}{dY_5} &= -\frac{1}{2} \cdot 2 \cdot (Ob_5 - Y_5) = -Err_5 \\
\frac{dLoss}{dY_6} &= -\frac{1}{2} \cdot 2 \cdot (Ob_6 - Y_6) = -Err_6 \\
\frac{dLoss}{dY_7} &= -\frac{1}{2} \cdot 2 \cdot (Ob_7 - Y_7) = -Err_7
\end{aligned}$$

Write the expressions for output nodes

$$\begin{aligned}
Y_5 &= Relu(W_{35}Y_3 + W_{45}Y_4 + b_5) \\
Y_6 &= Relu(W_{36}Y_3 + W_{46}Y_4 + b_6) \\
Y_7 &= Relu(W_{37}Y_3 + W_{47}Y_4 + b_7)
\end{aligned}$$

Now you can differentiate loss function with respect to all the output layer parameters $W_{35}, W_{45}, W_{36}, W_{46}, W_{37}, W_{47}, b_5, b_6, b_7$ using chain rule

$$\begin{aligned}
\frac{dLoss}{dW_{35}} &= \frac{dloss}{dY_5} \cdot \frac{dY_5}{dW_{35}} = -Err_5 \cdot Y_3 \cdot Relu'(W_{35}Y_3 + W_{45}Y_4 + b_5) \\
\frac{dLoss}{dW_{45}} &= \frac{dloss}{dY_5} \cdot \frac{dY_5}{dW_{45}} = -Err_5 \cdot Y_4 \cdot Relu'(W_{35}Y_3 + W_{45}Y_4 + b_5) \\
\frac{dLoss}{dW_{36}} &= \frac{dloss}{dY_6} \cdot \frac{dY_6}{dW_{36}} = -Err_6 \cdot Y_3 \cdot Relu'(W_{36}Y_3 + W_{46}Y_4 + b_6) \\
\frac{dLoss}{dW_{46}} &= \frac{dloss}{dY_6} \cdot \frac{dY_6}{dW_{46}} = -Err_6 \cdot Y_4 \cdot Relu'(W_{36}Y_3 + W_{46}Y_4 + b_6) \\
\frac{dLoss}{dW_{37}} &= \frac{dloss}{dY_7} \cdot \frac{dY_7}{dW_{37}} = -Err_7 \cdot Y_3 \cdot Relu'(W_{37}Y_3 + W_{47}Y_4 + b_7) \\
\frac{dLoss}{dW_{47}} &= \frac{dloss}{dY_7} \cdot \frac{dY_7}{dW_{47}} = -Err_7 \cdot Y_4 \cdot Relu'(W_{37}Y_3 + W_{47}Y_4 + b_7) \\
\frac{dLoss}{db_5} &= \frac{dloss}{dY_5} \cdot \frac{dY_5}{db_5} = -Err_5 \cdot Relu'(W_{35}Y_3 + W_{45}Y_4 + b_5) \\
\frac{dLoss}{db_6} &= \frac{dloss}{dY_6} \cdot \frac{dY_6}{db_6} = -Err_6 \cdot Relu'(W_{36}Y_3 + W_{46}Y_4 + b_6)
\end{aligned}$$

$$\frac{dLoss}{db_7} = \frac{dloss}{dY_7} \cdot \frac{dY_7}{db_7} = -Err_7 \cdot Relu'(W_{37}Y_3 + W_{47}Y_4 + b_7)$$

Now using gradient descent approach the update equations for all the output layer parameters are:

$$W_{35} = W_{35} + \eta Y_3 \cdot Err_5 \cdot Relu'(W_{35}Y_3 + W_{45}Y_4 + b_5) = \eta \cdot Y_3 \cdot \delta_5$$

$$W_{45} = W_{45} + \eta Y_4 \cdot Err_5 \cdot Relu'(W_{35}Y_3 + W_{45}Y_4 + b_5) = \eta \cdot Y_4 \cdot \delta_5$$

$$W_{36} = W_{36} + \eta Y_3 \cdot Err_6 \cdot Relu'(W_{36}Y_3 + W_{46}Y_4 + b_6) = \eta \cdot Y_3 \cdot \delta_6$$

$$W_{46} = W_{46} + \eta Y_4 \cdot Err_6 \cdot Relu'(W_{36}Y_3 + W_{46}Y_4 + b_6) = \eta \cdot Y_4 \cdot \delta_6$$

$$W_{37} = W_{37} + \eta Y_3 \cdot Err_7 \cdot Relu'(W_{37}Y_3 + W_{47}Y_4 + b_7) = \eta \cdot Y_3 \cdot \delta_7$$

$$W_{47} = W_{47} + \eta Y_4 \cdot Err_7 \cdot Relu'(W_{37}Y_3 + W_{47}Y_4 + b_7) = \eta \cdot Y_4 \cdot \delta_7$$

$$b_5 = b_5 + \eta Err_5 \cdot Relu'(W_{35}Y_3 + W_{45}Y_4 + b_5) = \eta \cdot \delta_5$$

$$b_6 = b_6 + \eta Err_6 \cdot Relu'(W_{36}Y_3 + W_{46}Y_4 + b_6) = \eta \cdot \delta_6$$

$$b_7 = b_7 + \eta Err_7 \cdot Relu'(W_{37}Y_3 + W_{47}Y_4 + b_7) = \eta \cdot \delta_7$$

Now consider the hidden nodes 3 and 4. Write the expressions for Y_3 and Y_4 .

$$Y_3 = Relu\{W_{13} \cdot X_1 + W_{23} \cdot X_2 + b_3\}$$

$$Y_4 = Relu\{W_{14} \cdot X_1 + W_{24} \cdot X_2 + b_4\}$$

Now differentiate the Loss function with respect to hidden outputs Y_3 and Y_4 . Both nodes 3 and 4 have parents 5, 6 and 7. Following the chain rule

$$\frac{dLoss}{dY_3} = \frac{dloss}{dY_5} \cdot \frac{dY_5}{dY_3} + \frac{dloss}{dY_6} \cdot \frac{dY_6}{dY_3} + \frac{dloss}{dY_7} \cdot \frac{dY_7}{dY_3} = (-\delta_5) \cdot W_{35} + (-\delta_6) \cdot W_{36} + (-\delta_7) \cdot W_{37}$$

$$\frac{dLoss}{dY_4} = \frac{dloss}{dY_5} \cdot \frac{dY_5}{dY_4} + \frac{dloss}{dY_6} \cdot \frac{dY_6}{dY_4} + \frac{dloss}{dY_7} \cdot \frac{dY_7}{dY_4} = (-\delta_5) \cdot W_{45} + (-\delta_6) \cdot W_{46} + (-\delta_7) \cdot W_{47}$$

Now we can differentiate Loss function with respect to all the hidden layer parameters W_{13} , W_{23} , W_{14} , W_{24} , b_3 and b_4

$$\frac{dLoss}{dW_{13}} = \frac{dLoss}{dY_3} \cdot \frac{dY_3}{dW_{13}} = \{(-\delta_5).W_{35} + (-\delta_6).W_{36} + (-\delta_7).W_{37}\} \cdot Relu'(W_{13}.X_1 + W_{23}.X_2 + b_3).X_1$$

$$\frac{dLoss}{dW_{23}} = \frac{dLoss}{dY_3} \cdot \frac{dY_3}{dW_{23}} = \{(-\delta_5).W_{35} + (-\delta_6).W_{36} + (-\delta_7).W_{37}\} \cdot Relu'(W_{13}.X_1 + W_{23}.X_2 + b_3).X_1$$

$$\frac{dLoss}{dW_{14}} = \frac{dLoss}{dY_4} \cdot \frac{dY_4}{dW_{14}} = \{(-\delta_5).W_{45} + (-\delta_6).W_{46} + (-\delta_7).W_{47}\} \cdot Relu'(W_{14}.X_1 + W_{24}.X_2 + b_4).X_1$$

$$\frac{dLoss}{dW_{24}} = \frac{dLoss}{dY_4} \cdot \frac{dY_4}{dW_{24}} = \{(-\delta_5).W_{45} + (-\delta_6).W_{46} + (-\delta_7).W_{47}\} \cdot Relu'(W_{14}.X_1 + W_{24}.X_2 + b_4).X_2$$

$$\frac{dLoss}{db_3} = \frac{dLoss}{dY_3} \cdot \frac{dY_3}{db_3} = \{(-\delta_5).W_{35} + (-\delta_6).W_{36} + (-\delta_7).W_{37}\} \cdot Relu'(W_{13}.X_1 + W_{23}.X_2 + b_3)$$

$$\frac{dLoss}{db_4} = \frac{dLoss}{dY_4} \cdot \frac{dY_4}{db_4} = \{(-\delta_5).W_{45} + (-\delta_6).W_{46} + (-\delta_7).W_{47}\} \cdot Relu'(W_{14}.X_1 + W_{24}.X_2 + b_4)$$

Therefore the updated equations for all the hidden layer parameters are as follows;

$$W_{13} = W_{13} + \eta \cdot (\delta_5.W_{35} + \delta_6.W_{36} + \delta_7.W_{37}) \cdot Relu'(W_{13}.X_1 + W_{23}.X_2 + b_3).X_1$$

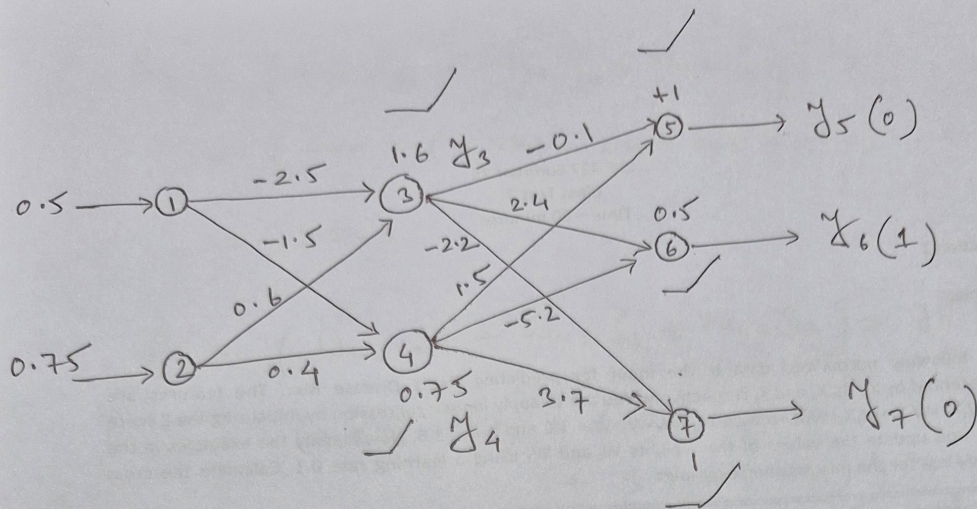
$$W_{23} = W_{23} + \eta \cdot (\delta_5.W_{35} + \delta_6.W_{36} + \delta_7.W_{37}) \cdot Relu'(W_{13}.X_1 + W_{23}.X_2 + b_3).X_2$$

$$W_{14} = W_{14} + \eta \cdot (\delta_5.W_{45} + \delta_6.W_{46} + \delta_7.W_{47}) \cdot Relu'(W_{14}.X_1 + W_{24}.X_2 + b_4).X_1$$

$$W_{24} = W_{24} + \eta \cdot (\delta_5.W_{45} + \delta_6.W_{46} + \delta_7.W_{47}) \cdot Relu'(W_{14}.X_1 + W_{24}.X_2 + b_4).X_2$$

$$b_3 = b_3 + \eta \cdot (\delta_5.W_{35} + \delta_6.W_{36} + \delta_7.W_{37}) \cdot Relu'(W_{13}.X_1 + W_{23}.X_2 + b_3)$$

$$b_4 = b_4 + \eta \cdot (\delta_5.W_{45} + \delta_6.W_{46} + \delta_7.W_{47}) \cdot Relu'(W_{14}.X_1 + W_{24}.X_2 + b_4)$$



$$y_3 = \text{Relu}(-0.5 \times 2.5 + 0.75 \times 0.6 + 1.6) = 0.8$$

$$y_4 = \text{Relu}(-0.5 \times 1.5 + 0.75 \times 0.4 + 0.7) = 0.3$$

$$y_5 = \text{Relu}(-0.1 \times 0.8 + 0.3 \times 1.5 + 1) = 1.37$$

$$y_6 = \text{Relu}(0.8 \times 2.4 - 0.3 \times 5.2 + 0.5) = 0.86$$

$$y_7 = \text{Relu}(-2.2 \times 0.8 + 0.3 \times 3.7 + 1) = 0.35$$

$$\delta_5 = \text{Err}_5 \cdot y_5' = (0 - 1.37) \times 1 = -1.37$$

$$\delta_6 = \text{Err}_6 \cdot y_6' = (1 - 0.86) \times 1 = 0.14$$

$$\delta_7 = \text{Err}_7 \cdot y_7' = (0 - 0.35) \times 1 = -0.35$$

$$\begin{aligned} \delta_3 &= (w_{35} \delta_5 + w_{36} \delta_6 + w_{37} \delta_7) y_3' \\ &= (-0.1 \times (-1.37) + 2.4 \times 0.14 + (-2.2) \times (-0.35)) \times 1 \\ &= 1.243 \end{aligned}$$

$$\begin{aligned} \delta_4 &= (w_{45} \delta_5 + w_{46} \delta_6 + w_{47} \delta_7) \times y_4' \\ &= (1.5 \times (-1.37) + (-5.2) \times 0.14 \\ &\quad + 3.7 \times (-0.35)) \times 1 \\ &= 4.078 \end{aligned}$$

$$w_{35} = w_{35} + \eta y_3 \delta_5 = -0.1 + 0.1 \times 0.8 \times (-1.37)$$

$$w_{36} = w_{36} + \eta y_3 \delta_6 = 2.4 + 0.1 \times 0.8 \times 0.14$$

$$w_{45} = w_{45} + \eta y_4 \delta_5 \quad w_{37} = w_{37} + \eta y_3 \delta_7$$

$$w_{46} = w_{46} + \eta y_4 \delta_6 \quad w_{47} = w_{47} + \eta y_4 \delta_7$$

$$b_5 = b_5 + \eta \delta_5 \quad b_6 = b_6 + \eta \delta_6 \quad b_7 = b_7 + \eta \delta_7$$

$$w_{13} = w_{13} + \eta x_1 \delta_3 = -2.5 + 0.1 \times 0.5 \times 1.243 = -2.4$$

$$w_{14} = w_{14} + \eta x_1 \delta_4 = -1.5 + 0.1 \times 0.5 \times 4.078 = -1.3$$

$$w_{23} = w_{23} + \eta x_2 \delta_3 \quad w_{24} = w_{24} + \eta x_2 \delta_4$$

$$b_3 = b_3 + \eta \delta_3 \quad b_4 = b_4 + \eta \delta_4$$